

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.3410626180240234, median 0.3018703756964685, std: 0.20670581819968228
Reprojection error (cam1): mean 0.29030860229796335, median 0.26168198516725505, std: 0.17024635427558432
Gyroscope error (imu0): mean 0.02781427295199618, median 0.021708779926194174, std: 0.021253626389721298
Accelerometer error (imu0): mean 0.029977895132646707, median 0.021681395941099156, std: 0.028287263328700924

Residuals

Reprojection error (cam0) [px]: mean 0.3410626180240234, median 0.3018703756964685, std: 0.20670581819968228
Reprojection error (cam1) [px]: mean 0.29030860229796335, median 0.26168198516725505, std: 0.17024635427558432
Gyroscope error (imu0) [rad/s]: mean 1.1525681254526647, median 0.8995686433573435, std: 0.8807079864841231
Accelerometer error (imu0) [m/s^2]: mean 0.1806037724819782, median 0.1306209752990167, std: 0.17041845158738003

Transformation (cam0):

T_ci: (imu0 to cam0):
[[0.99947394 -0.02727557 -0.01754689 0.08155904]
[0.0274498 0.99957547 0.00976631 0.00835312]
[0.01727306 -0.01024283 0.99979834 0.00170266]
[0. 0. 0. 1.]]

T_ic: (cam0 to imu0):
[[0.99947394 0.0274498 0.01727306 -0.08177484]
[-0.02727557 0.99957547 -0.01024283 -0.00610756]
[-0.01754689 0.00976631 0.99979834 -0.00035279]
[0. 0. 0. 1.]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
0.00478308376665382

Transformation (cam1):

T_ci: (imu0 to cam1):

```
[[ 0.9994284 -0.02683701 -0.02055842 0.00737699]
 [ 0.02702174 0.99959645 0.00876151 0.00812133]
 [ 0.02031499 -0.00931202 0.99975026 0.00124645]
 [ 0.         0.         0.         1.         ]]
```

T_ic: (cam1 to imu0):

```
[[ 0.9994284  0.02702174 0.02031499 -0.00761755]
 [-0.02683701 0.99959645 -0.00931202 -0.00790847]
 [-0.02055842 0.00876151 0.99975026 -0.00116564]
 [ 0.         0.         0.         1.         ]]
```

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
0.005060284378311684

Baselines:

Baseline (cam0 to cam1):

```
[[ 0.99999537 0.00040772 -0.0030162 -0.07417994]
 [-0.00041077 0.9999994 -0.0010122 -0.00019656]
 [ 0.00301579 0.00101344 0.99999494 -0.00071063]
 [ 0.         0.         0.         1.         ]]
```

baseline norm: 0.07418360667264533 [m]

Gravity vector in target coords: [m/s^2]
[9.80617586 -0.0781317 -0.03511992]

Calibration configuration

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cam0

Camera model: pinhole
Focal length: [401.608628861003, 399.428391132301]
Principal point: [319.7343936841441, 190.34744354361882]
Distortion model: radtan
Distortion coefficients: [-0.0076233519570338935, -0.0026557069530887645, -0.0008795023429080058, -0.00016680927611684287]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.03432 [m]
Spacing 0.010296000000000001 [m]

cam1

Camera model: pinhole
Focal length: [402.27012127979714, 400.2659622399602]
Principal point: [320.98190171333647, 190.80434036162683]
Distortion model: radtan
Distortion coefficients: [-0.0127486230660343, 0.009669430813801899, -0.0005299371877853309, 0.00021110128372654215]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.03432 [m]
Spacing 0.010296000000000001 [m]

IMU configuration

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IMU0:

Model: scale-misalignment
Update rate: 200

Accelerometer:

Noise density: 0.426001064

Noise density (discrete): 6.024564822941689

Random walk: 0.0010064709204016253

Gyroscope:

Noise density: 2.930109799

Noise density (discrete): 41.43801016988103

Random walk: 0.9815000742673874

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

[0. 1. 0. 0.]

[0. 0. 1. 0.]

[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

Gyroscope:

M:

[[0.00743805 0. 0.]

[-0.01308557 0.01302921 0.]

[0.0138178 -0.0200018 0.00889584]]

A [(rad/s)/(m/s²)]:

[[0.00753661 0.00507661 0.007183]

[-0.00352236 0.01435553 -0.0232746]

[0.02088948 0.00898483 -0.00251955]]

C_gyro_i:

[[0.89599886 -0.43477532 -0.0903132]

[0.32814636 0.7853071 -0.52498831]

[0.29917556 0.44075298 0.8463042]]

Accelerometer:

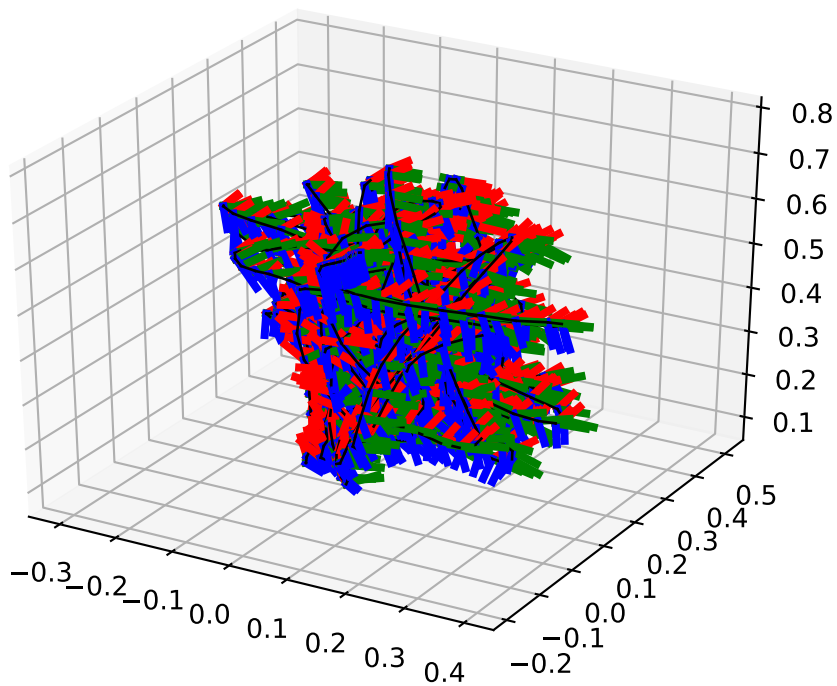
M:

[[0.98731977 0. 0.]

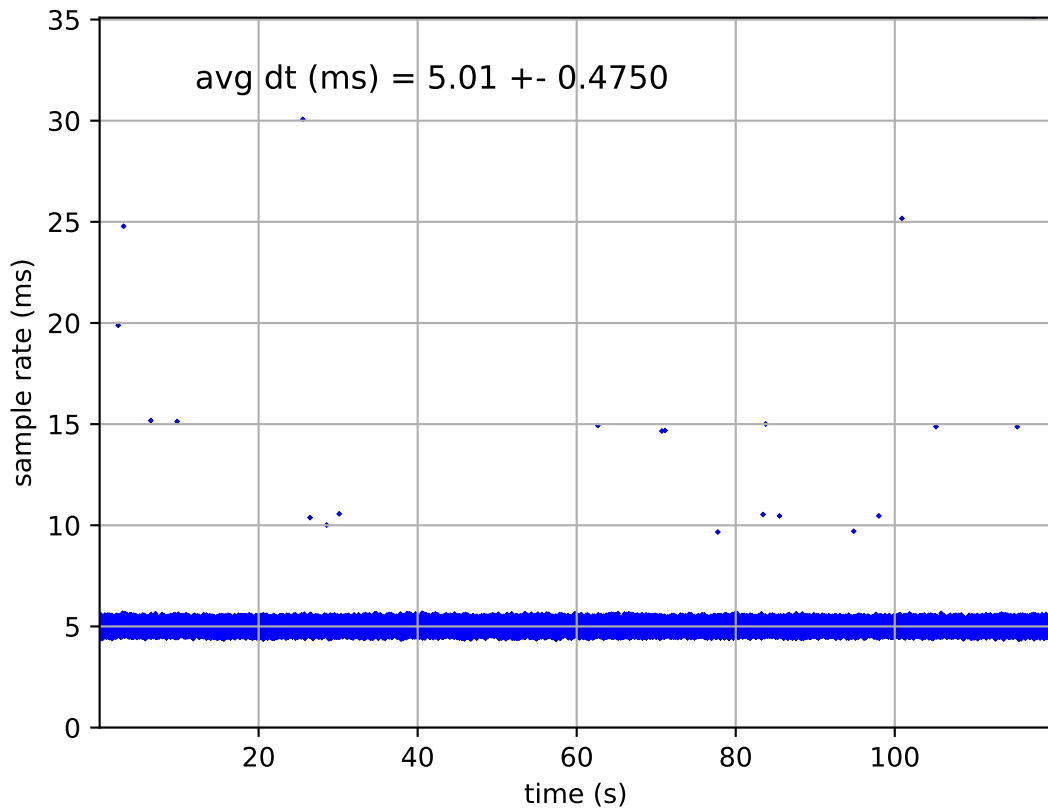
[0.00671096 0.9641481 0.]

[0.02194463 -0.01972741 0.98559633]]

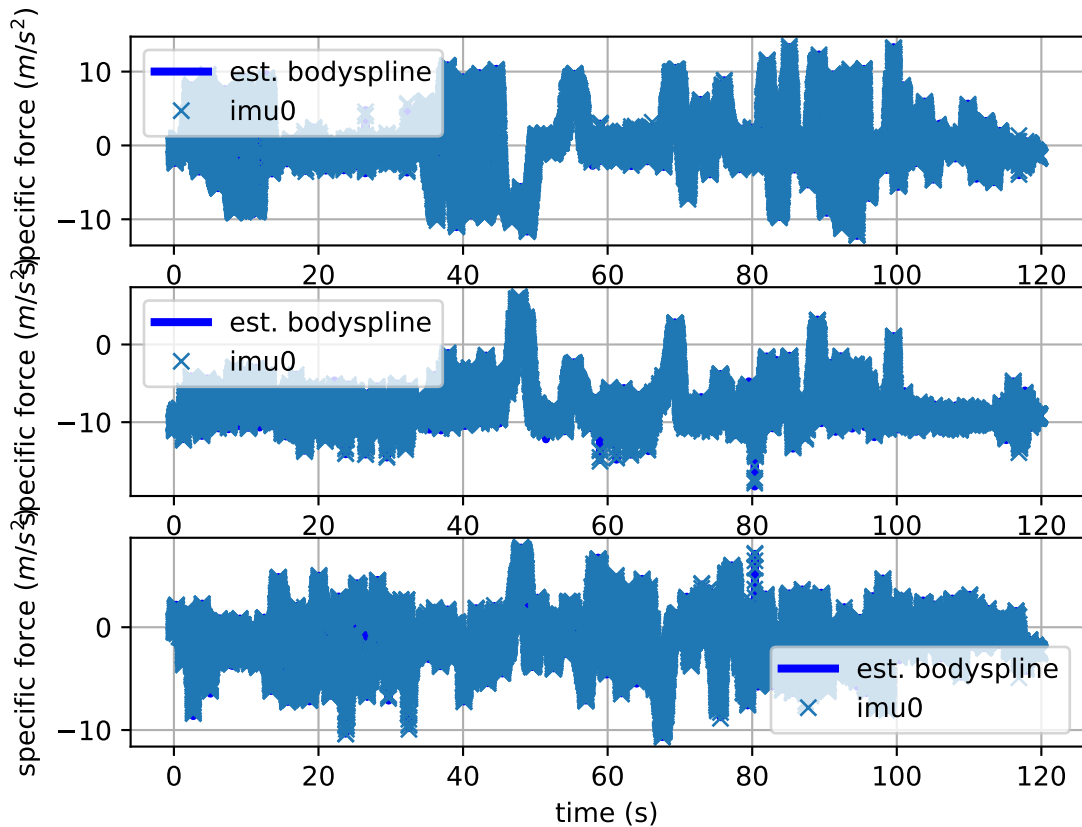
imu0: estimated poses



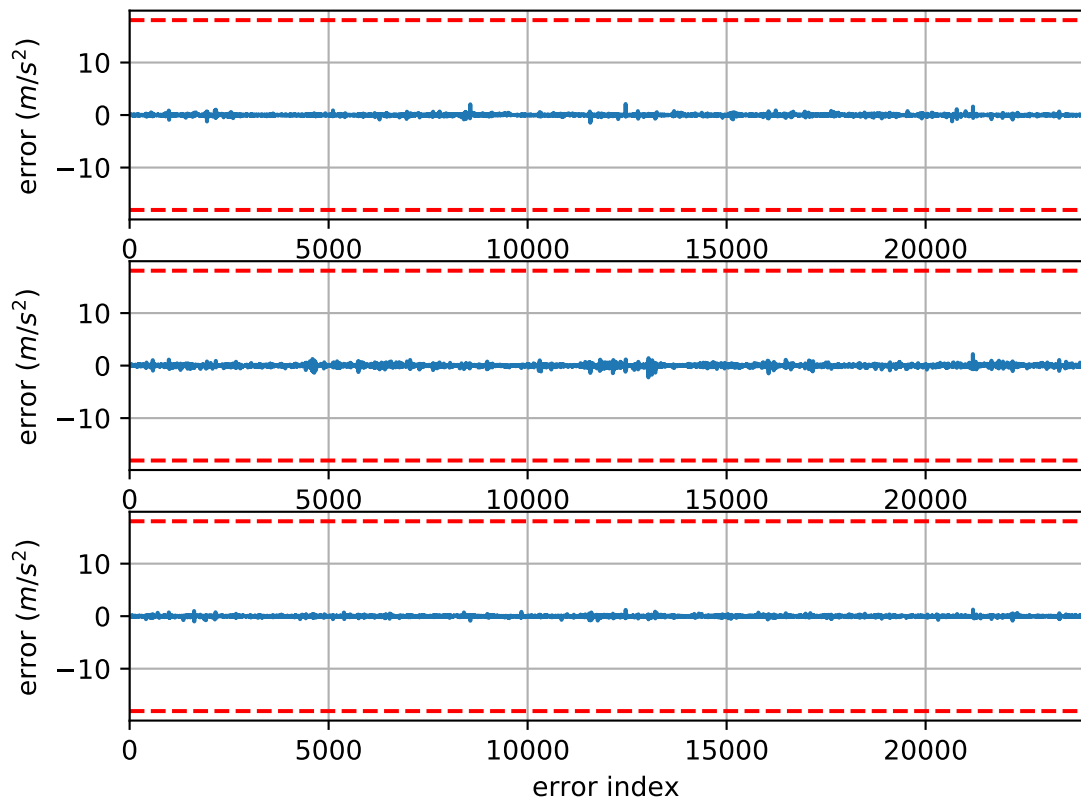
imu0: sample inertial rate



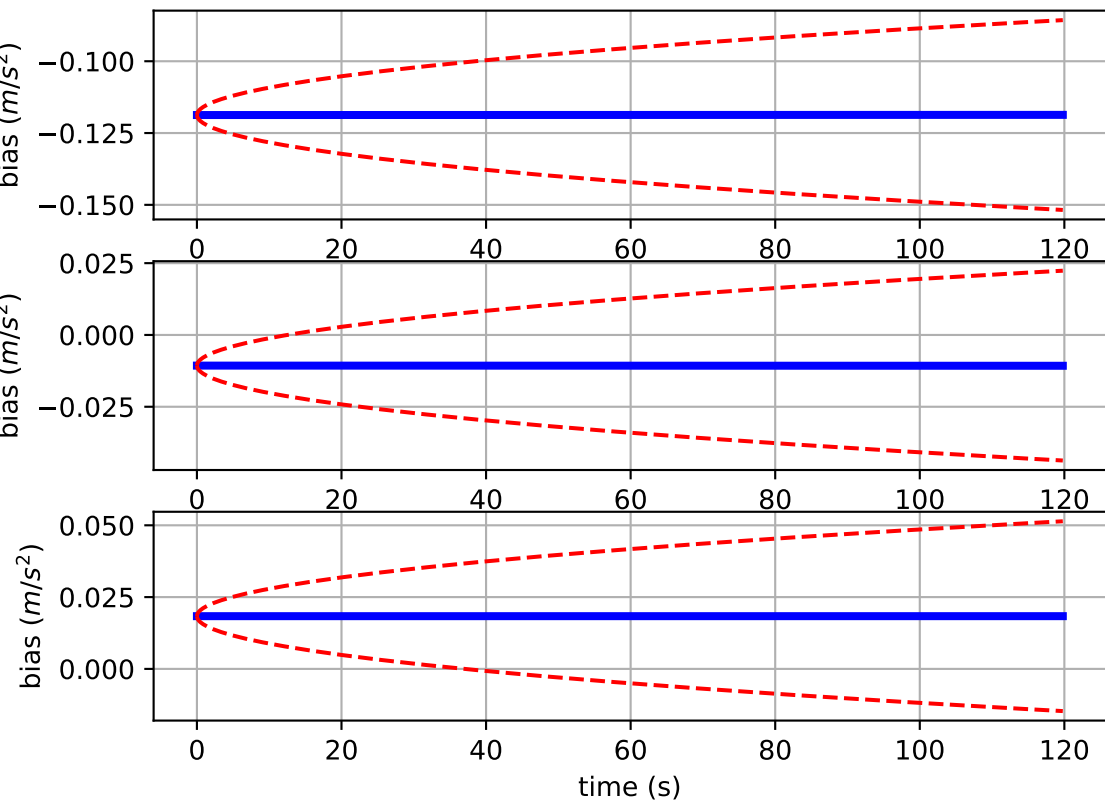
Comparison of predicted and measured specific force (imu0 frame)



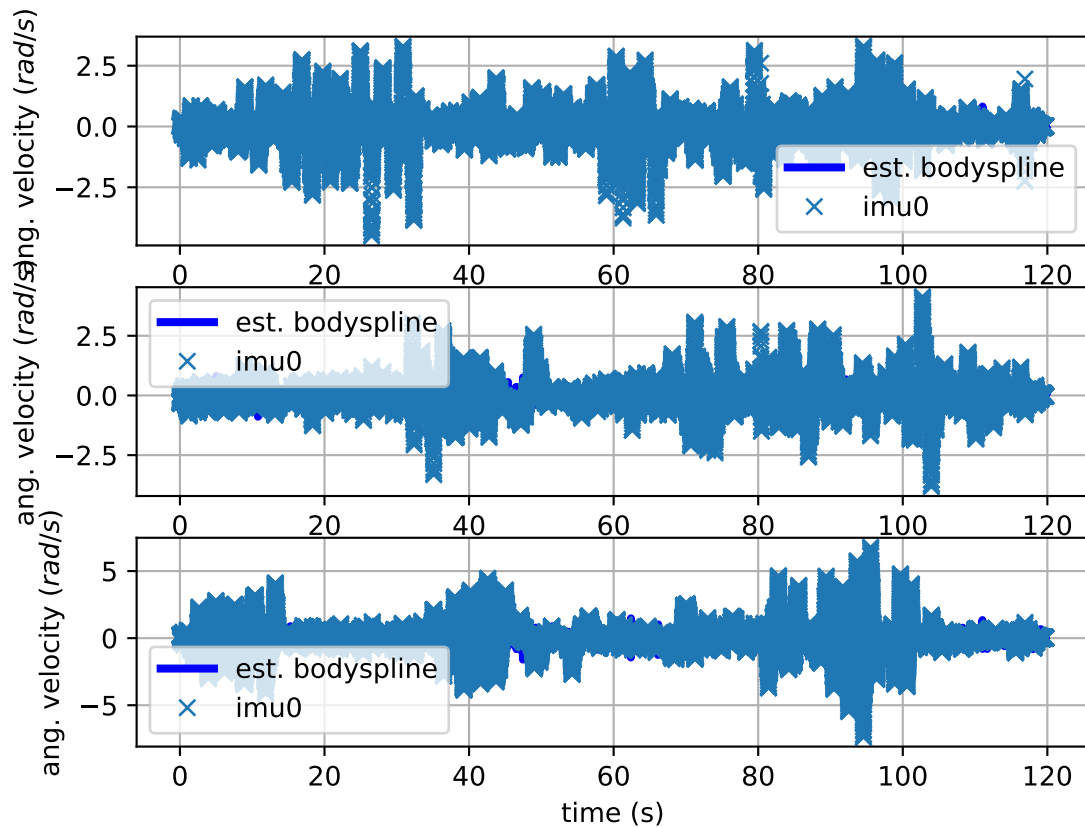
imu0: acceleration error



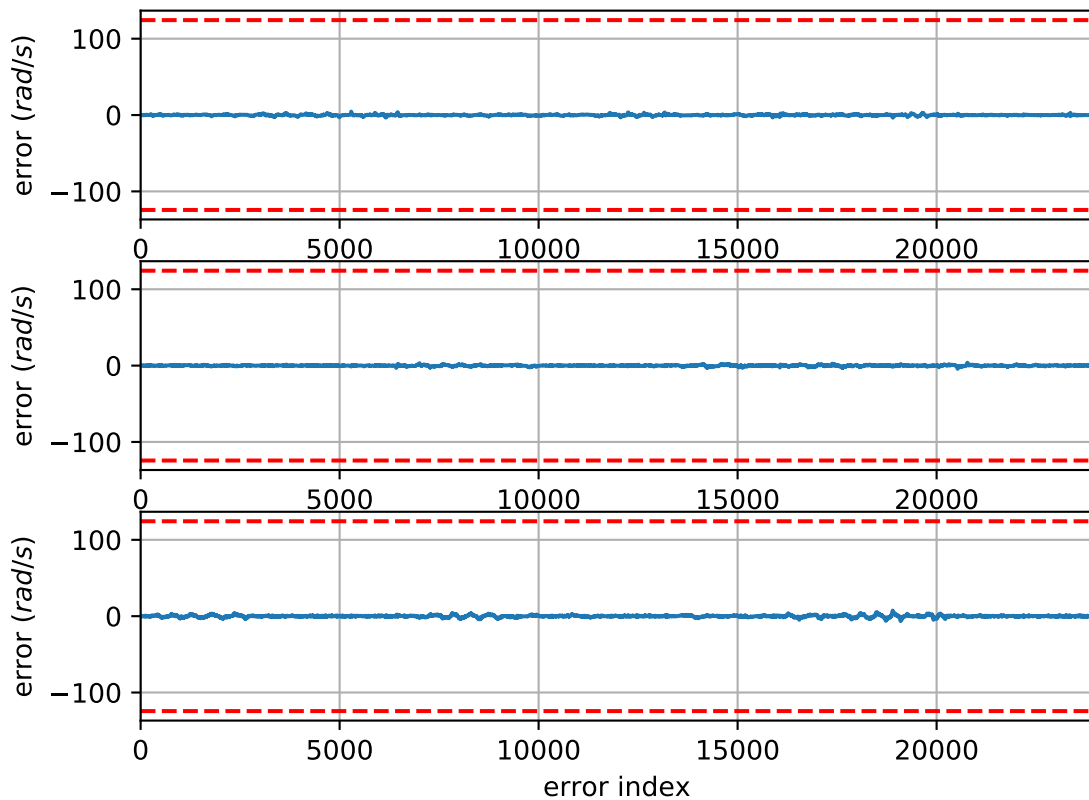
imu0: estimated accelerometer bias (imu frame)



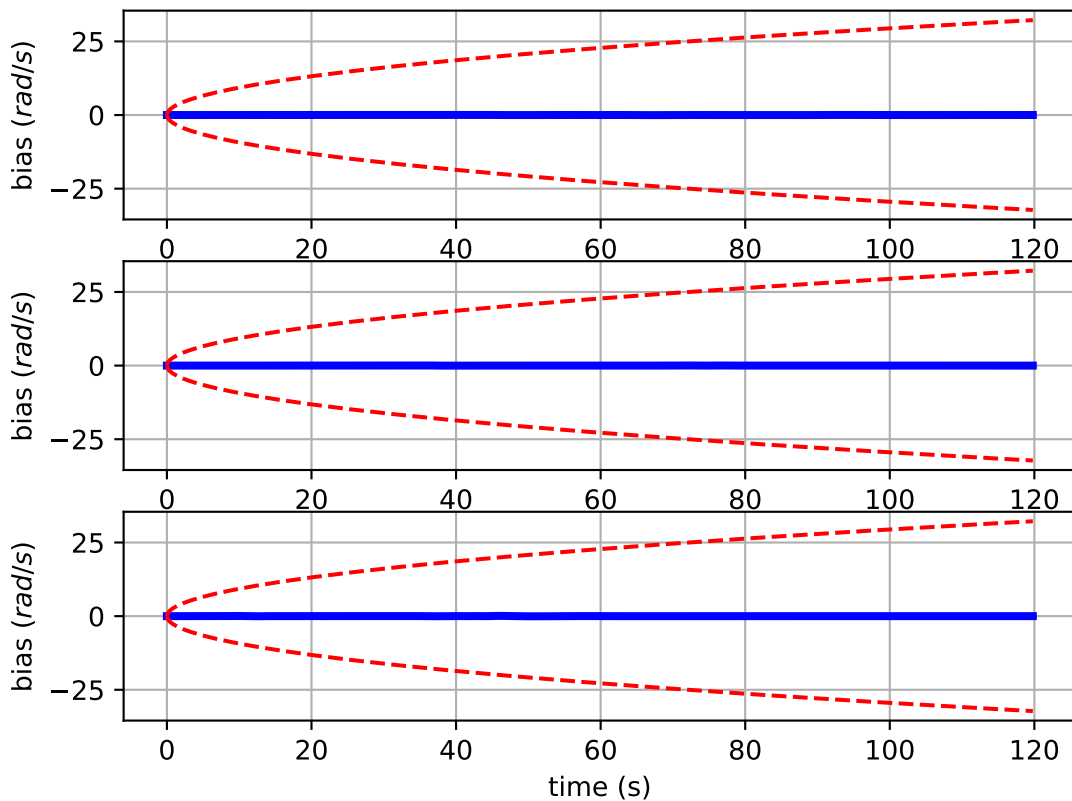
Comparison of predicted and measured angular velocities (body frame)



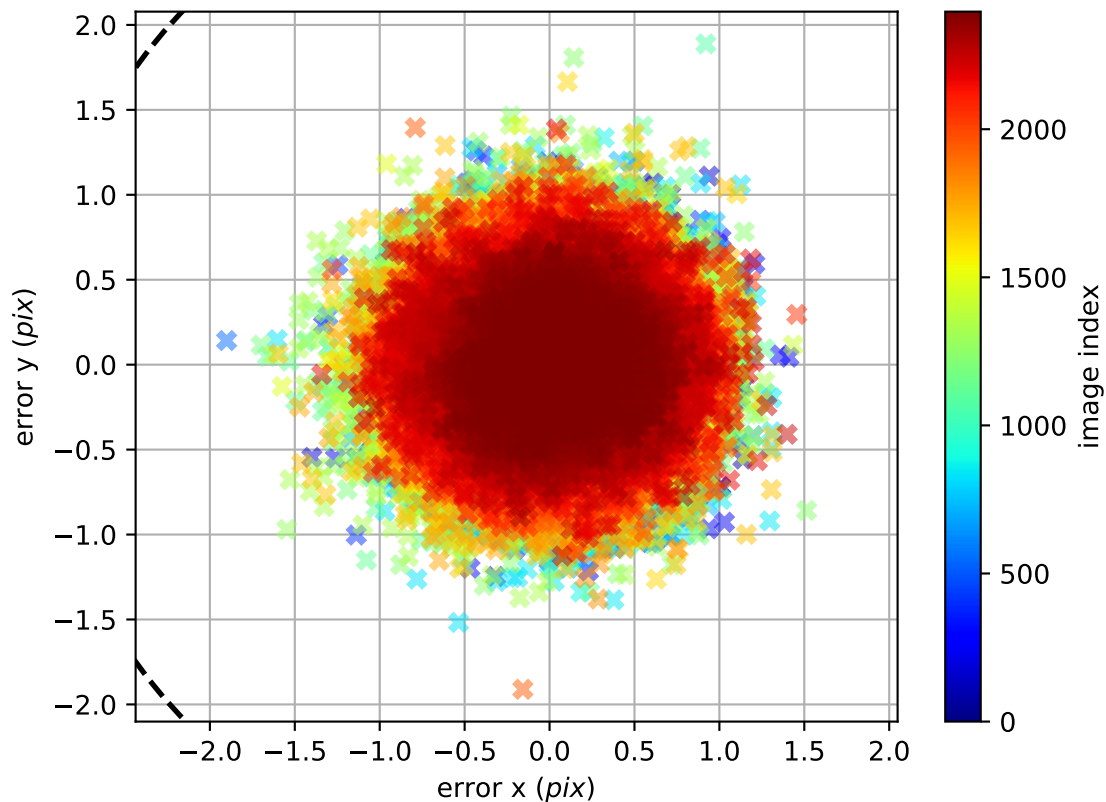
imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors



cam1: reprojection errors

